

## 1. Data Center

### Definition

A **Data Center** is a physical facility that contains computer systems, servers, storage devices, networking equipment, and cooling systems. It stores, processes, and manages data and applications.

### Components of a Data Center

- **Servers** – Process and store data.
- **Storage** – Stores files, databases, and backups.
- **Networking** – Routers, switches, and firewalls connect systems.
- **Power Supply** – Provides uninterrupted power (UPS and generators).
- **Cooling System** – Prevents servers from overheating.
- **Security** – CCTV, biometric access, and fire protection.

### Uses of a Data Center

- Store business data.
- Host websites and applications.
- Run cloud services.
- Database management.
- Backup and disaster recovery.

### Benefits

- High availability.
- Better security.
- Fast data access.
- Reliable performance.
- Centralized data management.

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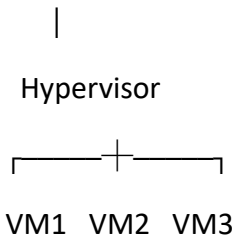
## 2. Virtualization

### Definition

**Virtualization** is the technology of creating multiple virtual machines (VMs) on a single physical computer using virtualization software.

### How it Works

Physical Server



Each VM has its own operating system and applications.

### Uses

- Run multiple operating systems.
- Improve server utilization.
- Testing and development.
- Cloud computing.
- Disaster recovery.

### Benefits

- Reduces hardware cost.
- Better resource utilization.
- Easy backup and migration.
- Faster deployment.
- Isolation between virtual machines.

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## 3. Hypervisor

### Definition

A **Hypervisor** is software that creates, manages, and runs Virtual Machines (VMs).

### Types of Hypervisor

#### Type 1 (Bare Metal Hypervisor)

- Installed directly on the physical hardware.
- No host operating system.
- High performance and better security.

#### **Examples**

- VMware ESXi
- Microsoft Hyper-V
- Xen

#### **Uses**

- Enterprise data centers.
  - Cloud providers.
  - Production servers.
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#### **Type 2 (Hosted Hypervisor)**

- Installed on top of an existing operating system.
- Easier to install.
- Mainly used for learning and testing.

#### **Examples**

- Oracle VirtualBox
- VMware Workstation

#### **Uses**

- Software testing.
  - Personal computers.
  - Development environments.
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#### **Difference Between Type 1 and Type 2 Hypervisor**

**Type 1****Type 2**

Runs directly on hardware      Runs on an operating system

Faster

Slightly slower

More secure

Less secure

Used in data centers

Used for testing and learning

Example: VMware ESXi

Example: VirtualBox

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**4. Features of Cloud Computing****1. On-Demand Self-Service**

Users can access resources whenever needed without human assistance.

**2. Broad Network Access**

Cloud services are accessible through the Internet from any device.

**3. Resource Pooling**

Resources are shared among multiple users using virtualization.

**4. Rapid Elasticity**

Resources can be increased or decreased quickly based on demand.

**5. Measured Service**

Users pay only for the resources they use (Pay-as-you-go).

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**5. Benefits of Cloud Computing**

- Lower cost.
- Easy scalability.
- High availability.
- Automatic updates.
- Data backup and recovery.

- Global accessibility.
  - Better collaboration.
  - Strong security.
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## 6. Vertical Scaling

### Definition

**Vertical Scaling (Scale Up/Down)** means increasing or decreasing the resources (CPU, RAM, Storage) of a **single server**.

### Example

A server with:

- 4 GB RAM → Upgrade to **16 GB RAM**
- 2 CPU → Upgrade to **8 CPU**

### Advantages

- Simple to implement.
- No application changes required.
- Better performance for one server.

### Disadvantages

- Limited by hardware capacity.
  - May require downtime.
  - More expensive after a certain limit.
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## 7. Horizontal Scaling

### Definition

**Horizontal Scaling (Scale Out/In)** means adding or removing **multiple servers** to handle increased or decreased workload.

### Example

Before:

Client

|

Server 1

After scaling:

Load Balancer

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Server1 Server2 Server3

### Advantages

- High availability.
- Better fault tolerance.
- Handles more users.
- Easy to expand.

### Disadvantages

- More complex to manage.
- Requires load balancer.
- Data synchronization may be needed.

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## Difference Between Vertical and Horizontal Scaling

### Vertical Scaling

### Horizontal Scaling

Add CPU/RAM to one server    Add more servers

Scale Up

Scale Out

Limited by hardware

Almost unlimited scalability

Easier to implement

More complex

May require downtime

Usually no downtime

Lower fault tolerance

Higher fault tolerance

- **Data Center:** A facility that houses servers, storage, networking devices, and other infrastructure to store, process, and manage data.
- **Virtualization:** The process of creating multiple virtual machines on a single physical server using a hypervisor.
- **Hypervisor:** Software that creates and manages virtual machines.
- **Vertical Scaling:** Increasing the resources (CPU, RAM, Storage) of a single server.
- **Horizontal Scaling:** Increasing capacity by adding more servers to distribute the workload.
- **Features of Cloud Computing:** On-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
- **Benefits of Cloud Computing:** Cost savings, scalability, high availability, security, backup and recovery, and global access.